

NEW PRODUCTS

MAKE IN INDIA

COMPONENTS

Automotive distance sensor

The new A19360 wheel speed and distance sensor is suitable for use in emerging applications in advanced



driver assistance systems (ADAS). The cutting-edge magnetoresistance (GMR) sensor

provides automakers with the signal resolution and reliability required for advanced levels of automation in passenger vehicles and mobility-as-a-service applications. It includes an eight-event-per-magnetic-cycle mode targeted at ADAS applications, providing an increment for every $\sim 5\text{mm}$ of tyre roll. It also includes a four-event-per-magnetic-cycle mode that doubles the number of outputs per magnetic cycle that allows designers to halve the number of poles on in-wheel ring magnets to save costs, or increase the air gap and still obtain the same number of increments per revolution.

Allegro MicroSystems
www.allegromicro.com

Low-current Zener diodes

These $50\mu\text{A}$ low-current voltage regulator Zener diodes are ideal for low bias and portable battery-powered devices in mobile, wearable, automotive, and



industrial applications. Available in three different surface-mountable

(SMD) package options, in an ultra-small discrete flat no-leads (DFN) package and AEC-Q101 qualified parts, they are the ultimate in customer choice and flexibility. The Zeners feature a non-repetitive peak reverse power dissipation of $\leq 40\text{W}$, total power dissipation $\leq 300\text{mW}$, and low dynamic resistance.

They are also available as Q-portfolio parts, meeting the AEC-Q101 and ISO/TS16949 automotive quality standards.

Nexperia
www.nexperia.com

MEMS solution

The XCalibur active resonator solves supply chain constraints using programmable semiconductors to deliver a drop-in replacement for quartz



crystal resonators. Additionally, XCalibur provides higher performance and reliability while reducing devel-

opment time by up to two months in automotive, enterprise, and industrial applications. XCalibur's active resonators provide 10x better reliability than quartz, making them ideal for robustness and long life in harsh environments. The many benefits of these MEMS based resonators also extend to communications and networking, enterprise, consumer electronics, mobile and IoT, and aerospace and defence applications.

SiTime
www.sitime.com

Buffer amplifier

The BUF802 is a wide-bandwidth high-input-impedance (Hi-Z) buffer amplifier, capable of supporting frequency



bandwidths as high as 3GHz . The wider

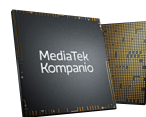
bandwidth and high slew rates enable higher signal throughput and minimal input settling time. By eliminating ASICs, designers who use this buffer can get to market faster while achieving a wide dynamic

range at a fraction of the cost. Designers can leverage this faster throughput to measure higher-frequency signals more accurately in test and measurement applications including oscilloscopes, active probes, and high-frequency data-acquisition systems.

Texas Instruments
www.ti.com

System-on-chip

The Kompanio 1380 chip offers a new level of performance and best-in-class features for premium Chromebooks,



such as the new Acer Chromebook Spin 513. The Kompanio 1380 delivers incredible

portable computing experiences with superb battery life, all in a compact form factor for slim and lightweight devices. Features include powerful processing with an octa-core CPU having four high-performance Arm Cortex-A78 cores with speeds up to 3GHz for improved responsiveness, powerful multi-core AI processor, $4\text{K}60\text{ HDR}$ display support, giving device makers a wide variety of resolution, performance, and external connectivity choices to meet any product design, AV1 hardware decoding, voice assistance, and ultra-fast connectivity.

MediaTek
www.mediatek.com

NFC Type 2 tag ICs

The ST25TN512 and ST25TN01K NFC Forum Type 2 tag ICs deliver a new



balance of cost and performance for high-volume use cases like consumer

engagement, product information, and brand protection. Also suitable for smart-city applications and access control, the ST25TN512/01K NFC tag ICs support multiple user-protection